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1 Appendix F: Plasmonic Nano-Geometry Sweep

1.1 Objective

Sweep nanoparticle radii and media to quantify resonance frequency, quality factor, and field enhancement relevant to receptor microstructures.

1.2 Planned Methods (src)

- `src/meta_sweep.py`
 - `sweep_plasmonic_quality(radii_m, metal_eps, medium_eps)`

1.3 Planned Script and Outputs

- Script: `scripts/generate_plasmonic_geometry_sweep.py`
- Data: `output/data/plasmonic_geometry.npz`
- Figure: `output/figures/plasmonic_geometry_sweep.png`

1.4 Figure

1.5 Equation References

- Resonance/wavelength: see plasmonic definitions in main text; material equations in ??.

1.6 Reproducibility

- Run: `python3 scripts/generate_plasmonic_geometry_sweep.py`
- Artifacts saved to `output/data/` and `output/figures/`.

1.7 Cross-References

- Methods: ??
- Symbols: ??
- Math appendix: ??

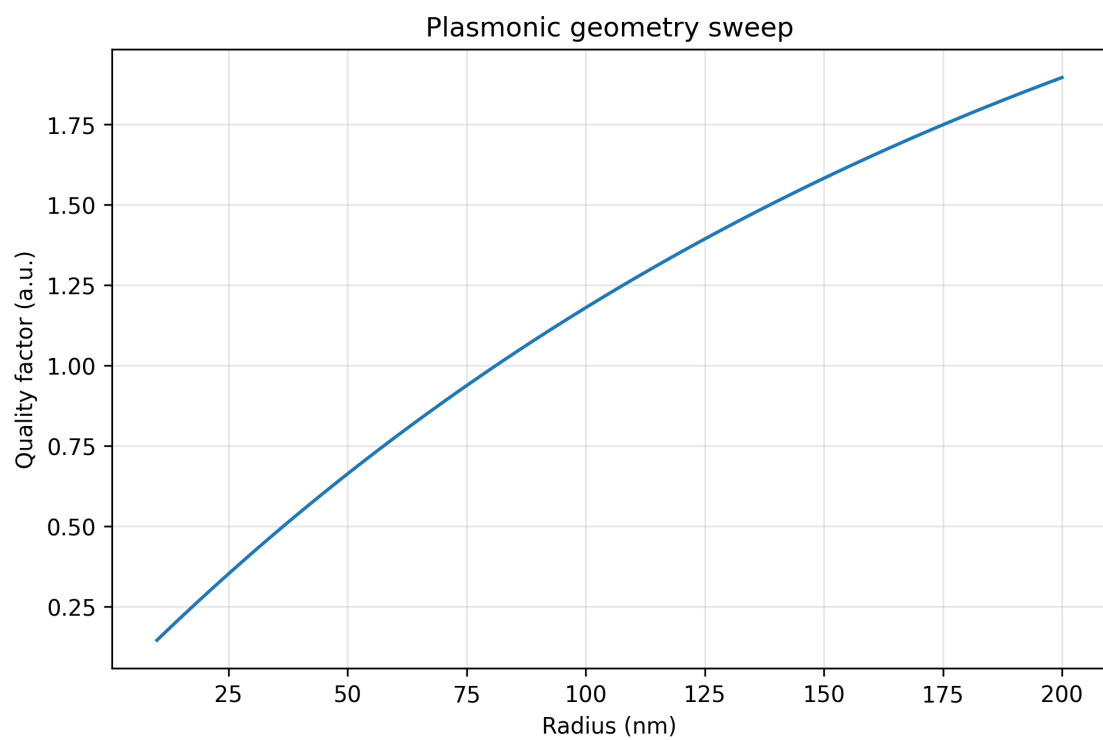


Figure 1: *Resonance frequency, Q factor, and field enhancement as functions of nanoparticle radius and medium dielectric.*